

BOOKEXCHANGE

Connecting Students. Saving Money. Reusing Books.

ENSE 281 | Scrum 2 Presentation| March 18, 2026

QR Code:



Meet the Team

Thomas P.

GitHub Administrator
Frontend Developer

Chop Peter

Project Manager

David

Lead Developer

Shaikh

QA Lead
Documentation

The Problem

Students spend **\$1,200+/year** on textbooks

Campus bookstore buyback: only **10-20%** of original price

No campus-specific platform exists to connect student buyers and sellers

Result:

- Buyers pay full price
- Sellers get almost nothing back
- Books sit unused or are discarded

Our Solution

A web application where students buy and sell used textbooks directly to each other



**Buyers save
30-70%**



**Sellers get
fair value**



**Books get
reused**

Chop (Project Manager)

What he did:

Defined the core MVPs (Post, Search, Connect) and kept the team focused on a functional loop rather than getting distracted by unnecessary features (like a complex in-app chat).

How he went about it:

Broke the project down into an iterative process. First, get the mock database running. Second, build the UI. Third, connect the dynamic routing

The Challenge

Coordinating testing when the app was only running locally on David's machine, which created a bottleneck for the designer (Sami) and frontend developer (Thomas) to review changes

The Solution:

Implemented localtunnel to create public URLs bridging to local environments. Ran into 503 gateway error initially, but diagnosed it as a port mismatch (e.g. conflicting with past projects running on port 3000 vs 3001) and established a strict "two-terminal" rule for the team to test the live server securely.

Thomas (GitHub Administrator, Frontend Developer)

What he did:

Built out the user-facing forms and integrated the EJS (Embedded JavaScript) templating engine to generate the HTML dynamically based on what David's backend sent over

How he went about it:

Created the advanced search sidebar, swapping out rigid radio buttons for dynamic `<select>` dropdowns for university departments, and built the "Empty State" UI components for when queries return zero results.

The Challenge

Flexbox alignment issues across different pages. The navigation menu kept breaking and shoving all links to the far right on pages like "My Listings" and "Search" making the app look disjointed.

The Solution:

Standardized the HTML structure. Extracted the navigation into a universal modular `<nav>` block with `.nav-left` and `.nav-right` containers. This guaranteed the flexbox layout remained perfectly consistent no matter which EJS view the user loaded.

David (Lead Developer)

What he did:

Architected the [Node.js/Express](#) server, set up Multer for image file uploads, and engineered the secure “double-blind” email relay system using Nodemailer.

How he went about it:

Used local JSON files (books.json, users.json, messages.json) to mock a database for the MVP. Built custom filter functions to process Thomas’ frontend search queries (handling case insensitivity and empty fields).

The Challenge

Two major hurdles. First, the server kept crashing randomly with a [SyntaxError: Unexpected end of JSON input]. Second, the professor’s requirement to protect user privacy meant we couldn’t just display seller emails on the frontend.

The Solution: * The Crash Fix:

Realized Node’s [JSON.parse()] crashes when reading a completely empty 0-byte file. Wrote a robust try/catch wrapper that checks if(!data.trim()) and safely returns an empty array to prevent fatal server crashes.

The Privacy Fix: Built a middleman relay. Instead of exposing emails, the server generates a unique UUID for a conversation. Nodemailer emails the seller a secure localhost:3000/reply/UUID link. When they click, they reply via the website.

Shaikh(QA Lead, Documentation)

What he did:

Designed the application to feel authentic to the student body by utilizing the official U of R deep forest green (#005030) and vibrant gold color palette, alongside a clean, modern card-based layout for the textbooks.

How he went about it:

Focused on a “dashboard” feel. Ensured the “Call to Action” buttons (like the red Delete button) contrasted heavily with the primary navigation to prevent accidental clicks.

The Challenge

Integrating the custom SVG BookExchange logo without distorting the strict height constraints of the navigation bar, and dealing with layout shifts where the search sidebar would float to the center of the screen instead of staying at the top.

The Solution:

Used CSS transformations (`transform: scale()`) anchored to the left to visually enlarge the logo without breaking the physical DOM space. Fixed the search layout by implementing strict flexbox rules (`align-items: flex-start`) to force the sidebar and results grid to the top of the viewport

Self-Reflection

How did you feel about this activity?

This activity was both challenging and rewarding. It pushed our team to turn an idea into a working software prototype and helped us experience the realities of software development under time and resource constraints.

What did you like about it?

We liked that the project resulted in something tangible and demonstrable. It was satisfying to see the prototype support the main marketplace workflow, including browsing books, posting listings, viewing details, contacting sellers by email, and managing personal listings.

What did you dislike about it?

The most difficult part was that we had to make trade-offs because of time, complexity, and budget limitations. We could not implement every desired feature in this phase, and running the project only through localhost limited how realistic the deployment experience could be.

"We learned that good documentation and planning upfront saves time later in development."

Self-Reflection (Continued)

What is your team most proud of?

We are most proud of building a working prototype that demonstrates the core purpose of the platform. Even though the system is not yet fully complete, it successfully shows the main user journey and gives us a strong base for future improvements.

What did you learn about yourself as you collaborated and worked on this activity?

Individually, team members learned how to better manage their time under pressure and discovered their personal strengths in either front-end design or back-end logic. We learned that collaboration works best when responsibilities are clear and the team agrees early on what is realistic. We also learned that reducing scope is sometimes the best decision if it helps deliver a stable and credible prototype.

How will you use what you learned going forward?

Going forward, we will use this experience to plan projects more strategically, define scope earlier, and separate essential features from optional enhancements. We will also pay more attention to deployment planning and technical feasibility from the start.

What "stuff & things" related to this activity would you like more help with?

We would like more help with secure authentication, deployment and hosting decisions, and designing future features such as in-app messaging and peer rating systems.

Next Steps

- **Replace the partial login flow with a secure authentication system.**
- **Add validation, stronger error handling, and cleaner UI feedback.**
- **Consider in-app messaging as a future enhancement.**
- **Consider a peer rating or review system to improve trust and safety.**
- **Deploy the application when hosting resources become available.**

Thank You!

Questions?

GitHub: github.com/thomaskeithpersson-afk/281-Group-Project